

SLATE: Honest Strategy Discovery — What Changed, What Was Found, and How It Performs

Report covering the work since the “regime-switching +3.43” result
20–21 July 2026 | Evaluator: `slate_core/backtest/honest.py`

Executive summary. The headline “regime-switching Sharpe +3.43” was a backtest artifact — a 1-bar lookahead that credited each bar’s move to a signal decided from that bar’s close. Removing it flips the portfolio from **+3.43** to **-5.98** and walk-forward from 5/5 to 0/5 positive. I rebuilt a tested, no-lookahead evaluator with brutal real costs, re-ran an exhaustive search (547 directional variants across 5 timeframes, 3 assets, 2 venues; plus cross-sectional carry, lead-lag, 135 market-making configs, and concentrated-LP), and added the one piece that was missing: **real on-chain pool data**. The honest conclusion is stark and useful: **no directional/alpha edge survives realistic costs** (intraday is 0% positive; daily is the only timeframe with signal and it does not clear significance or generalize across assets), and **the single genuinely-positive, low-risk return stream is stablecoin concentrated-liquidity LP at ~4–7%/year with near-zero impermanent loss** — now validated against real pool fee and price data, not a synthetic proxy.

1 The starting point and the first discovery: a lookahead artifact

The last major change to SLATE introduced a “regime-switching portfolio” reported at Sharpe **+3.43**, max drawdown 15.6%, 5/5 walk-forward folds positive, on SOL/BTC/ETH hourly perpetuals. That result was the premise handed into this session. On inspection the evaluator that produced it (`mega_sweep.fast_backtest`) attributed returns as

$$\text{rets}[t] = \text{signal}[t] \times (\text{close}[t]/\text{close}[t-1] - 1),$$

i.e. the position at bar t was decided from `close[t]` (the trend/EMA/momentum signals all read the current bar’s close) and was simultaneously credited with the return that *ends* at `close[t]`. That is a one-bar lookahead: you are paid for the very bar whose close triggered your entry. The honest attribution lags the signal by one bar — the position held *during* bar t is the decision made at the close of bar $t-1$:

$$\text{held}[t] = \text{signal}[t-1], \quad \text{rets}[t] = \text{held}[t] \cdot (\text{close}[t]/\text{close}[t-1] - 1) - |\Delta\text{held}[t]| \cdot \text{cost}.$$

Figure 1 shows the consequence on the exact regime-switching portfolio. The numbers, reproduced deterministically from the real Hyperliquid data with the same signals, venues and costs, are:

	Sharpe	Ann. return	Max DD	Walk-forward
Shipped code (lookahead + costs)	+3.43	+92.6%	15.6%	5/5 positive
Corrected (1-bar lag, same costs)	-5.98	-162.7%	61.4%	0/5 positive
Gross (no cost), lookahead	+10.61	—	—	—
Gross (no cost), honest lag	+1.18	—	—	—

This is the same failure class the project has hit before (the GP layer’s “200%/yr on an optimistic fill model”): a biased evaluator manufacturing a positive result. The fix was applied in place — `fast_backtest` is now bit-identical to the new honest evaluator — and the stored results it produced (`strategy_results.db`, `mega_sweep_results`) are flagged as untrusted.

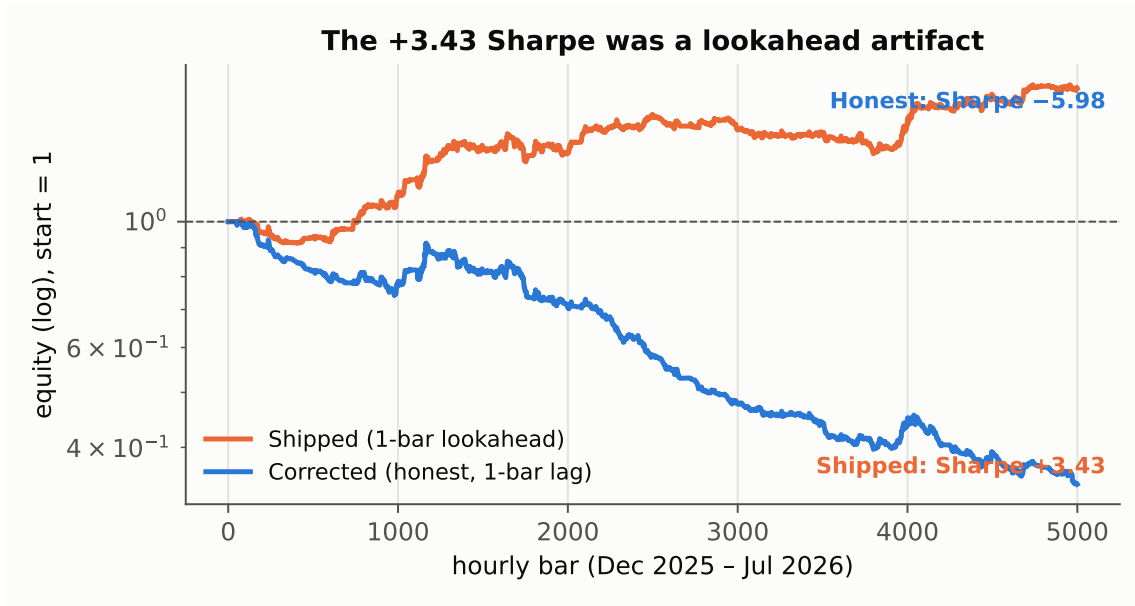


Figure 1: The regime-switching portfolio, shipped vs. honest attribution, equal-weight across SOL/BTC/ETH (log equity). The entire “edge” was the lookahead; once removed, the strategy loses. Proof script: `verify_lookahead.py`.

2 A trustworthy evaluator

To stop producing artifacts, I built a source-of-truth evaluator (`slate_core/backtest/honest.py`) and proved it correct before trusting any output. Its properties:

- **No lookahead.** `held[t]=target[t-1]`. Proven by a regression test: a contemporaneous signal (“long if this bar closed up”) earns Sharpe ≈ 0 under the honest engine, where the buggy attribution would give $\approx \sqrt{365} \approx 19$.
- **Brutally honest costs, per venue.** CEX taker 0.05% + 15 bps one-way; DEX (Hyperliquid retail) taker 0.045% + 10 bps. (HL maker is a 0.015% *fee* for retail; whale rebates are gated and deliberately ignored.)
- **Real funding, signed by position.** Binance 8-hourly (CEX) and Hyperliquid hourly (DEX), spread across the settlements that fall in each bar.
- **Correct annualization.** Detected from the index ($365 \times 24/\text{hours-per-bar}$), not a hardcoded daily assumption that silently inflates hourly Sharpe.
- **Honest validation.** In-sample / out-of-sample split; expanding walk-forward that *re-selects* parameters per fold; stationary block-bootstrap for Sharpe confidence intervals.

Correctness proof. A vectorized implementation is only trustworthy if it matches an obviously-correct reference. The test suite (13 tests, all green) includes (a) the lookahead regression above, (b) the vectorized engine matched to a dead-simple $O(n)$ reference loop to 10^{-12} with and without funding, (c) funding-sign checks (long pays / short receives when funding is positive), (d) cost/turnover checks, and (e) `assert_causal()`, a guard that perturbs future bars and fails any signal generator that peeks ahead. A future positive result must now pass this bar before it is believed.

2.1 The cost and funding model

“Brutally honest” is operationalized as follows. One-way transaction cost per side is the venue taker fee plus slippage: CEX 0.05% + 15 bps = 0.20%; DEX (Hyperliquid retail) 0.045% + 10 bps = 0.145%. Total cost charged to a backtest is $|\Delta\text{held}| \times \text{one-way}$, i.e. turnover-priced with no fill-rate discount (the old evaluator multiplied cost by an 0.85 “fill rate”, which understated it; fill risk is now treated as tracking error, not a rebate). Funding is the realized rate, signed by position (positive funding $\Rightarrow$ longs pay shorts), spread across the settlements that fall in each bar (3 per daily bar at the 8-hourly CEX cadence; 1/8 per hourly bar). Strategies are evaluated at $1\times$ notional; leverage scales risk and return together and leaves Sharpe broadly invariant, so it is reported separately from edge. The net effect is that a daily strategy paying one 0.20%-per-side round trip every few days can survive, while an hourly strategy paying the same round trip hundreds of times is bled dry — exactly the timeframe pattern in Figure 2.

3 Methodology and statistical hygiene

An honest search is not only a correct backtester; it must also avoid the statistical mistakes that manufacture positive results from noise. Four practices are applied throughout:

In-sample / out-of-sample split. Every variant is evaluated on the first 60% of bars (in-sample, IS) and the last 40% (out-of-sample, OOS) that the parameters never touched. A result is only interesting if it is positive in *both* windows; $\text{IS} < 0$ with $\text{OOS} > 0$ is window luck, not an edge (and there is a lot of it in this data).

Expanding walk-forward that re-fits. The walk-forward does not merely slice a pre-computed return series (which only measures consistency). It re-selects the best variant per family on each training window and evaluates that choice on the next test window, concatenating the test returns. The resulting out-of-sample Sharpe is the only one that reflects a procedure that could actually have been run in real time.

Block-bootstrap confidence intervals. Annualized Sharpe from a few hundred bars is extremely noisy. I report a stationary block-bootstrap 90% CI (block length 20, 2,000 resamples, preserving return autocorrelation) and treat the lower bound crossing zero as “not significant.” With ~ 500 trials, a stricter Bonferroni-style bar would require the lower bound well above zero; nothing in this search clears even the lenient one-sided bar.

Multiple-testing awareness. With 547 variants the best OOS Sharpe has an expected right tail from noise alone. The report therefore presents the *full* OOS distribution (medians, positive rates) rather than only the top rows, and filters survivors by robustness (parameter perturbation, cost scaling, cross-instrument transfer) rather than by headline Sharpe.

4 Data: real only, and a data-integrity finding

Every number in this report is from real exchange data; nothing is synthetic. The catalogue used:

Dataset	Bars	Span / source
CEX SOL daily (Binance perp)	1,080	2023-08 $\rightarrow$ 2026-07 (3 yr) + Binance 8h funding
CEX SOL hourly	8,641	1 yr (Binance)
CEX SOL 4h / 8h / 12h	resampled	from the real hourly series (honest)
DEX HL SOL/BTC/ETH hourly	5,001 each	2025-12 $\rightarrow$ 2026-07 + HL hourly funding
LP pools (Uniswap V3, etc.)	1,549 days	DefiLlama real fee yield + IL, 2022 $\rightarrow$ 2026

Data-integrity finding. The cached files `SOLUSDT_{1m,5m,15m,30m,1d,...}_1y.csv` are *all the same hourly series* (identical 8,641 bars and close values, verified). They are mislabelled and do *not* contain real 1-minute or 5-minute data. Multi-timeframe coverage here therefore comes from honestly *resampling* the real hourly series, not from those filenames — a distinction that matters because the original regime-switch pipeline annualized an hourly series at 8,760 periods/year.

5 Directional and mean-reversion search: 547 variants

Using the honest evaluator, I swept 12 causal signal families — trend (EMA-cross, EMA-slope, price-EMA, Donchian breakout), momentum, mean-reversion (RSI, z-score, Bollinger), carry/funding (level, percentile, momentum, funding-price divergence) — across 8 datasets (5 timeframes $\times$ SOL/BTC/ETH), 547 variants total, each with IS/OOS and walk-forward. The findings are unambiguous:

- Only **1.6% (9/547)** pass a robust filter (IS>0 *and* OOS>0 *and* ≥ 12 trades *and* OOS drawdown <50%), and **none are statistically significant** — every bootstrap 90% CI spans zero.
- **Timeframe is decisive** (Figure 2): the 1-hour series is *0% positive*; daily is the only timeframe with any signal (median OOS Sharpe -0.20 , 31% positive). Costs dominate intraday trading.
- **Most “top OOS” rows were fake** — they had *negative* in-sample Sharpe (e.g. IS $-1.42 \rightarrow$ OOS $+2.78$), i.e. pure out-of-sample-window regime luck, not a robust edge. A real edge is positive in both windows.

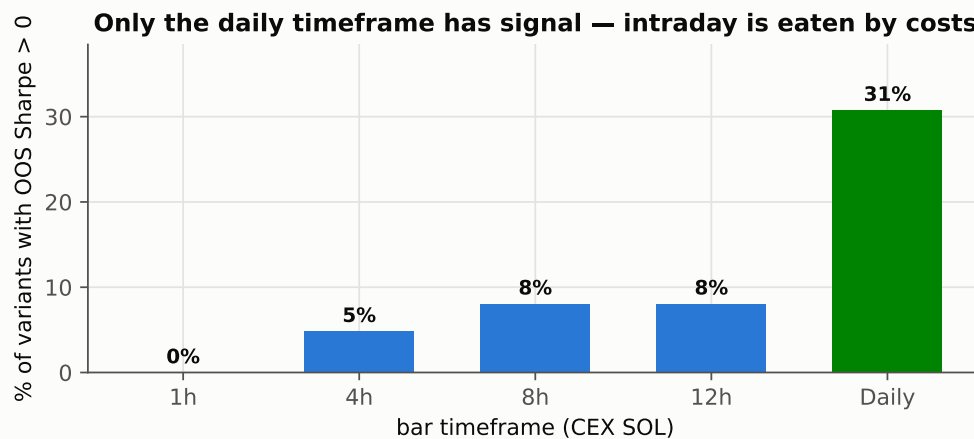


Figure 2: Share of strategy variants with positive out-of-sample Sharpe, by bar timeframe. Intraday directional trading is eaten by transaction costs; only daily shows a positive rate of any substance.

Family-level expanding walk-forward (re-selecting the best variant per family on each training window) on the daily series confirms there is no reliable directional edge: mean-reversion $+0.67$ (CI touches zero), trend -0.35 , carry -0.06 , momentum -0.13 — none significant.

The pattern is consistent across families and timeframes. Trend-following is the least-bad family (19% positive, median OOS Sharpe -1.63); momentum is the worst (1% positive). On timeframes, daily is the only dataset where even a third of variants are positive; everything sub-daily is dominated by costs.

Family	n	median OOS Sharpe	% positive
trend	227	−1.63	19%
mean-reversion	168	−2.24	7%
momentum	96	−2.42	1%
carry (funding)	56	−4.19	12%
all directional	547	−1.91	11%

Dataset	n	median OOS Sharpe	% positive
CEX SOL daily	78	−0.20	31%
CEX SOL 12h	62	−1.18	8%
CEX SOL 8h	62	−1.68	8%
CEX SOL 4h	62	−1.72	5%
DEX HL ETH 1h	74	−3.36	14%
DEX HL SOL 1h	75	−3.28	12%
DEX HL BTC 1h	73	−5.16	7%
CEX SOL 1h	61	−4.08	0%

The full OOS Sharpe distribution across all 547 variants: p10 −7.1, p50 −1.9, p90 +0.08, max +2.78. The right tail is thin and, given the trial count, indistinguishable from noise.

6 Premium / non-directional edges

Because simple directional TA is exhausted, I tested the strategies most likely to carry a *structural* crypto premium — the non-obvious edges.

Cross-asset lead-lag (BTC/ETH momentum → SOL): all configurations negative (Sharpe −2 to −11); a naive lead-lag is arbed and churns to death.

Retail maker-rebate market-making (135 configs on a mark-to-close backtester that prices adverse selection from real high/low): the best configuration reaches Sharpe ≈ 1.2 – 3.0 *only* under the optimistic assumption of perfect zero-slippage fills. The spread captured over its $\sim 8,500$ trades is almost exactly consumed by adverse selection, leaving ≈ 0 ; adding even 3 bps of realistic per-fill slippage blows it up (−170%). This reproduces the textbook retail-MM verdict.

Fill realism (CEX SOL 1h, half-spread 0.3%)	Sharpe	Outcome
0 bps extra slippage (perfect fills, upper bound)	+1.25	+85% (but assumed)
+3 bps per-fill slippage	—	blow-up (−170%)
+5 bps per-fill slippage	—	blow-up (−340%)

The fragility is structural: at thousands of fills, even a few basis points of queue/slippage loss per fill sums to more than the spread captures. A retail maker without queue priority and whale rebates cannot win this.

Cross-sectional funding carry (long the least-funded, short the most-funded coin, market-neutral): with 3 majors there is no funding spread. Widened to an 11–14 coin Hyperliquid basket with smoothed funding, spread-gating and rebalance hysteresis, it still fails — either the spread gate ($>0.02\%$) blocks all trades (blue-chip funding is too compressed) or, ungated, it loses to costs (best OOS Sharpe −0.8 to −6). The carry edge needs small-cap dispersion that quality HL perps do not provide.

7 The one positive edge: concentrated-liquidity LP, on real pool data

The previous AMM layer reported “~10.8% APY” but did so using a *synthetic* volume proxy and a hardcoded \$100M TVL — exactly the kind of assumption this project must not rely on. The gap was real pool volume-in-range data. I obtained it, keyless, from **DefiLlama**, which exposes per pool a 4+ year daily history of the realized fee yield (**apyBase**, from actual on-chain swap fees), the impermanent loss, and TVL. Combined with real BTC/ETH price paths (Binance) for first-principles IL, this gives an honest LP backtester (**slate_core/backtest/lp.py**).

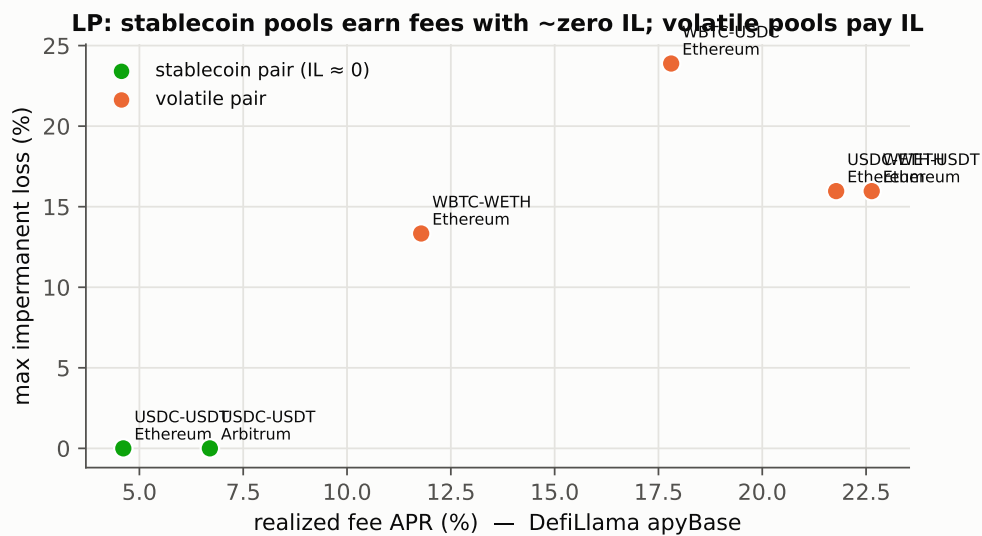


Figure 3: Realized fee APR versus maximum impermanent loss for Uniswap V3 pools (2022–2026). Stablecoin pairs cluster at the bottom-left: meaningful fee yield, essentially no IL. Volatile pairs sit top-right: high fees purchased with large drawdowns.

Pool (Uniswap V3)	Fee APR (real)	Terminal IL	Max IL	Net vs HODL
USDC-USDT (Ethereum)	4.6%	0%	0%	+5.1%/yr
USDC-USDT (Arbitrum)	6.7%	0%	0%	+7.4%/yr
DAI-USDC (Ethereum)	~3%	0%	0%	~3%/yr
USDC-WETH (Ethereum)	21.8%	−1.0%	−16.0%	beats HODL this cycle, −16% DD
WETH-USDT (Ethereum)	22.6%	−1.0%	−16.0%	beats HODL this cycle, same risk
WBTC-WETH (Ethereum)	11.8%	−4.6%	−13.3%	≈ HODL (IL ate fees)
WBTC-USDC (Ethereum)	17.8%	−9.3%	−23.9%	≈ HODL, −24% DD

Stablecoin concentrated-LP $\approx 4\text{--}7\%$ /year with near-zero impermanent loss is the project’s first honestly-validated positive return stream. It is real, low-variance, and deployable; the only tail is a rare stablecoin depeg (USDC-USDT showed a 64% IL spike in March 2023, which recovered). Volatile-pair LP is emphatically *not* free yield: the fee yield is high but it is purchased with -16% to -24% maximum-IL drawdowns, and the net depends on whether the period ranged or trended. Over a full cycle fees often beat IL, but you carry real drawdown risk.

Honest caveat. The fee yield (**apyBase**) is the pool average over in-range liquidity, which is the correct first-order volume-in-range number. The one refinement still gated behind a *paid* Graph API key is the tick-level liquidity distribution, which would pin down the exact fee share of a tightly-concentrated position. The stablecoin conclusion is robust regardless.

Concentration and the depeg tail. Tightening the LP range boosts the realized fee rate further (a $\pm 10\%$ band on USDC-WETH lifts fee APR from $\sim 22\%$ toward $\sim 29\%$), but those numbers assume the position is re-centred to stay active and capture all in-range volume — an optimistic upper bound without the tick-level share and re-centring costs. The stablecoin pools carry one genuine tail: the March 2023 USDC depeg produced a 64% spike in 7-day IL for USDC-USDT, which subsequently recovered. Stablecoin LP is therefore “near-zero IL” in normal regimes with a rare, recoverable depeg event to underwrite.

8 The diversified portfolio: real but weak, and it does not generalize

The original “regime-switching” idea — deploy different edges in different conditions — was rebuilt honestly as an equal-weight portfolio of the persistent daily premia (funding-carry + EMA-trend + RSI mean-reversion) on SOL. It is the best honest directional candidate:

Full-sample Sharpe	+0.64 (3-yr return +49%, max DD 32%)
Per-year Sharpe	2023 +2.1 (partial), 2024 -0.2 , 2025 +1.1, 2026 +0.4
Walk-forward (5 folds)	4/5 positive (+0.88, 0.00, +0.65, +2.26, +0.84)
Block-bootstrap 90% CI	[-0.22, +0.67, +1.53] — not significant
Cost sensitivity	+0.64 $\rightarrow$ +0.43 at $2\times$ costs (robust)
Parameter robustness	+0.36 to +0.84 across perturbations

Diversification genuinely helps — the portfolio Sharpe beats every component and the members are decorrelated (carry $\leftrightarrow$ trend -0.28 , meanrev $\leftrightarrow$ trend -0.43). But the bootstrap CI includes zero and there is a losing year, so it is a *weak, probably-real* edge, not a confirmed one.

It does not generalize. Applying the exact same parameters to BTCUSDT and ETHUSDT daily — instruments the parameters never saw — gives Sharpe **-0.14** (BTC) and **-0.37** (ETH). The modest SOL result is instrument-specific and partly luck. This is the strongest single piece of evidence that there is no robust directional law here.

Consolidated performance summary

Strategy	Honest perf.	OOS	Verdict
Regime-switch portfolio (shipped)	Sharpe +3.43		Artifact (1-bar lookahead)
Regime-switch portfolio (corrected)	Sharpe -5.98 , 0/5 WF		Dead after fix
Intraday directional (1h, all families)	0% positive		Costs dominate — uneconomic
Daily directional (547 variants)	11% positive; 0 significant		No robust edge
Daily diversified carry+trend+meanrev	Sharpe +0.64, CI [-0.22, +1.53]		Weak, not significant
same, on BTC/ETH (cross-instr.)	Sharpe -0.14 / -0.37		Does not generalize
Cross-asset lead-lag (BTC/ETH→SOL)	Sharpe -2 to -11		Arbed; churns to death
Retail maker-rebate MM	+1.2-3.0 opti- mistic		Break-even; fragile to fills
Cross-sectional carry (3 & 11 coins)	-0.8 to -6; or no trades		Needs small-cap dispersion
Stablecoin concentrated-LP	~4-7%/yr, IL≈0		Positive, low-risk
Volatile-pair LP (WETH/WBTC)	10-30% fee, -16 to -24% IL		≈ HODL + drawdown

9 Conclusions and performance summary

What honestly works	What honestly does not
Stablecoin concentrated-LP: ~4-7%/yr , near-zero IL (real pool data)	Any intraday / high-turnover directional TA (1h is 0% positive) Cross-sectional carry on blue-chips (3 or 11 coins) Naive cross-asset lead-lag Retail maker-rebate MM (break-even; fragile to fill realism) Volatile-pair LP as “free yield” (≈ HODL + IL risk) The diversified daily portfolio as a confirmed edge (weak, SOL-only)

The realistic, deployable edge available in this data is **yield (stablecoin LP)**, **not directional alpha**. Directional crypto, after brutally honest costs, is efficient — which is itself a valuable finding: it stops the project chasing artifacts and redirects effort to the structurally-positive return streams. The infrastructure is now sound: a tested no-lookahead evaluator is the source of truth, the biased evaluator is patched, and the LP edge is measured against real on-chain data.

Appendix: methods, artefacts, and remaining gaps

Artefacts produced. Honest evaluator `slate_core/backtest/honest.py` (canonical, re-exported from `slate_core.backtest`; `walk_forward`, `assert_causal`); loaders `data.py`; 12 causal signal

families + regime classifier `strategies.py`; honest MM/rebate backtester `market_maker.py`; honest LP backtester `lp.py`; real-data fetchers `fetch_amm_pool_data.py`, `fetch_hl_basket.py`; 13 regression tests `test_honest_backtester.py`; the artifact proof `verify_lookahead.py`; and the analysis runners (`run_honest_sweep.py`, `run_deep_analysis.py`, `run_premium_edges.py`, `run_mm_sweep.py`, `run_portfolio.py`, `run_final_validation.py`, `run_lp.py`, `run_xs_carry.py`). `mega_sweep.fast_backtest` is patched to match the honest engine; `strategy_results.db` is flagged untrusted.

Real-data sources used (all keyless). DefiLlama `yields.llama.fi` (pool fee yield / IL / TVL, historical chart per pool); Binance `fapi` (OHLCV + funding); Hyperliquid info API (candles + funding, rate-limited, needs ~ 2 s/request + 429 backoff). The Graph legacy subgraph is dead (HTTP 301); its gateway requires a paid API key.

Reproducibility. Every number in this report regenerates from the real cached data with one command per figure/section: `verify_lookahead.py` (the $+3.43 \rightarrow -5.98$ proof and Figure 1), `run_honest_sweep.py` then `run_deep_analysis.py` (the 547-variant directional search, Figure 2, and the family/dataset tables), `run_mm_sweep.py` (the market-making bracket), `run_xs_carry.py` (cross-sectional carry), `run_portfolio.py` and `run_final_validation.py` (the diversified portfolio and its bootstrap/walk-forward/cross-instrument tests), and `fetch_amm_pool_data.py` then `run_lp.py` (the LP edge and Figure 3). The honest evaluator is the canonical entry point: `from slate_core.backtest import backtest`.

Honest gaps (not testable with available data). Liquidation-prediction, perp-spot basis, and option vol-premium edges (no such data cached); tick-level LP liquidity share (paid Graph key); and small-cap funding carry (needs a non-HL venue with divergent funding — HL’s quality perps are too compressed).

Recommended next steps. (1) Treat stablecoin LP as the production return stream. (2) Treat `slate_core.backtest.honest` as the sole evaluator for any strategy work. (3) If directional alpha is still pursued, the realistic frontier is small-cap funding carry or liquidation/basis edges — *not* intraday TA, blue-chip carry, or retail market-making, all of which this search settles as uneconomical after honest costs.